Name: _____

WORKSHEET

# Solving periodic functions

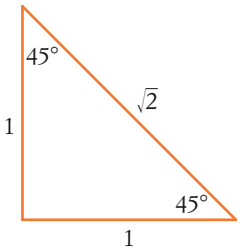
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| | | | | | | | | | | | |
| 11 | 8 | 6 | 10 | 6 | 3 | 11 | 11 | 8 | 12 | 9 | 5 | 3 |

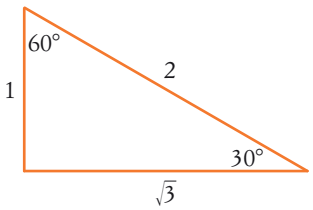
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|---|---|---|---|---|---|---|---|
| | | | | | | | |
| 12 | 9 | 1 | 12 | 4 | 6 | | |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |
| 7 | 2 | 6 | 9 | 11 | 11 | 8 | 12 | 9 | 5 | 3 | |

*Art Buchwald*

## Rules




## Questions

Simplify the following periodic functions, then use the above triangles to find exact answers.

**1** $\tan(180° + 30°)$

**2** $\cos\left(\pi + \frac{\pi}{4}\right)$

**3** $\sin\left(2\pi - \frac{\pi}{6}\right)$

**4** $\tan(180° + 60°)$

**5** $\sin\left(\frac{3\pi}{2} - \frac{\pi}{6}\right)$

**6** $\cos(360° - 60°)$

**7** $\tan\left(\pi - \frac{\pi}{4}\right)$

**8** $\tan\left(-\frac{\pi}{3}\right)$

**9** $\sin(90° + 45°)$

**10** $\tan\left(2\pi + \frac{\pi}{4}\right)$

**11** $\cos(270° + 60°)$

**12** $\tan\left(\pi - \frac{\pi}{6}\right)$

## Solutions

**A** $-1$

**B** $1$

**E** $\frac{1}{2}$

**F** $\sqrt{3}$

**G** $-\frac{\sqrt{3}}{2}$

**H** $-\sqrt{3}$

**I** $-\frac{1}{\sqrt{3}}$

**L** $\frac{1}{\sqrt{3}}$

**N** $\frac{1}{\sqrt{2}}$

**R** $-\frac{1}{\sqrt{2}}$

**S** $-\frac{1}{2}$

**T** $\frac{\sqrt{3}}{2}$

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